

Mohammad Ful Hossain Seikh

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Education

Ph.D. in Physics 2020–2026

University of Kansas, USA

3.964/4.0

M.S. in Electrical Engineering 2024–2025

University of Kansas, USA

3.778/4.0

M.Sc. in Physics 2018–2020

Aligarh Muslim University, India

9.170/10 (Gold Medalist)

B.Sc. Honors in Physics 2015–2018

Aligarh Muslim University, India

9.186/10 (Gold Medalist)

Research Experience

Research Roles and Appointments

- **Postdoctoral Researcher: Particle Astrophysics, University of Kansas** *Jun 2026–Present*
Research on ultra-high energy particle detection with radio techniques, including the ARA five-station data analysis, radio-background characterization, and detector studies with Prof. D. Besson.
- **Core Analyst, ARA Five-Station Diffuse Neutrino Search** 2022–Present
One of eight analyzers conducting the first array-wide diffuse ultra-high energy neutrino search using 10.6 years of ARA data. Integrated the ARA01 analysis into the common multi-station framework and contributed to array-wide optimization, unblinding, and interpretation of the final result.
- **Analysis Lead, Non-Anthropogenic Down-going Signals in ARA** *Jul 2026–Present*
Leading an array-wide search across all five ARA stations for down-going radio signals inconsistent with known anthropogenic backgrounds.
- **Analysis Lead, Radio Background Characterization with ARA** *Dec 2025–Present*
Leading studies of astrophysical, environmental, and anthropogenic radio backgrounds in ARA data, including solar activity, galactic noise, high-wind triboelectric events, radiosondes, and local sources.
- **Primary Analyzer, Askaryan Radio Array (ARA) Station 1** 2021–2026
Led the end-to-end ARA01 analysis for ultra-high energy neutrino searches, including detector calibration and data-driven modeling, event reconstruction, background rejection, and simulation. Developed the station-specific analysis framework later incorporated into the full-array search.
- **Monitoring Shifter, IceCube Neutrino Observatory** 2021–2026
Served as KU representative for monitoring shifts, presenting weekly reports to the collaboration.
- **Visiting Summer Student** *May–Jul 2019*
Studied the role of θ_{13} in three-flavor neutrino oscillations with Prof. S.K. Agarwalla at the Institute of Physics, India, producing technical report “Importance of 1-3 Mixing Angle in Three-Flavor Neutrino Oscillations Paradigm.” DOI: 10.13140/RG.2.2.34421.12009.
- **Winter Research Intern** *Dec 2019–Jan 2020*
Researched “Weak Interaction Physics” at Indian Institute of Technology–BHU with Prof. G. Abbas.

Selected Dissertations, Theses, and Projects

- **Physics PhD Dissertation** 2020–2026
“Toward a Five-Station Diffuse Ultra-High Energy Neutrino Search with the Askaryan Radio Array: ARA01 Analysis Framework.” Defended May 8, 2026. Advised by Prof. D. Besson. Published on ProQuest ISBN: 9798252480190 and KU ScholarWorks.

- **Electrical Engineering Master’s Project 2024–2025**
Developed **AAFIYA**, an open-source Python toolkit for antenna measurement, simulation, and characterization, with Prof. Jim Stiles at KU EECS. The project led to an accepted contribution at the 2026 IEEE AP-S/URSI Symposium. (GitHub), (Report), (Slides).
- **Machine Learning Course Project 2025**
Developed a CNN model for “Automatic Surface Detection in Echograms using Deep Learning” in a KU EECS machine-learning course. (GitHub), (Report).
- **Physics Master’s Thesis 2019–2020**
Investigated light sterile neutrino oscillations with Prof. M.S. Athar at Aligarh Muslim University, producing the review report “Light Sterile Neutrinos: Status and Prospects.” DOI:10.13140/RG.2.2.13777.66405.
- **Physics Undergraduate Project 2017–2018**
“Quarks & Leptons”: detailed study of Standard Model interactions with Prof. M.S. Athar at AMU (Handwritten Report).

Scientific Software and Open-Source Development

- **MAHAM: Modules for Astroparticle, High-energy, Astronomy and Multimessenger 2026–Present**
Founder and lead developer of **Project MAHAM**, an open-source Python toolkit under active development for reusable, validated, and citable scientific methods and data products in astroparticle physics, high-energy astrophysics, astronomy, and multimessenger research. Established the package architecture, scientific provenance and validation framework, with current development spanning neutrino spectra and models, experimental limits and sensitivities, detector-response calculations, statistical methods, and reproducible literature datasets.
- **AAFIYA: Antenna Analysis in Frequency-domain for Impedance & Yield Assessment 2024–Present**
Creator and developer of **AAFIYA**, an open-source Python toolkit for reproducible antenna characterization using measurement and simulation data. Implements S-parameter and impedance analysis, gain and beam-pattern characterization, beamwidth, front-to-back and cross-polarization metrics, realized vector effective length, and measurement-to-simulation validation, with ongoing development toward a more integrated reusable software package.

Field Experience

- **NSF’s Summit Station, Greenland Field Campaign Apr–May 2025**
Participated in the Radio Neutrino Observatory in Greenland (RNO-G) calibration campaign, performing in situ radio-frequency measurements and equipment deployment at Summit Station, Greenland.

Publications & Manuscripts

Manuscripts in Collaboration Review

Core analysis team: Bishop, A., Clark, B.A., Dasgupta, P., Giri, P., Machtay, A., Muzio, M.S., Salcedo-Gomez, A., and **Seikh, M.F.H.**

- **ARA Collaboration (2026).** “First Array-Wide Search for Ultrahigh Energy Neutrinos with the Askaryan Radio Array.” *Under Collaboration review; planned submission to Physical Review Letters.*
- **ARA Collaboration (2026).** “Search for Ultrahigh Energy Neutrinos using 10.6 Years of Askaryan Radio Array Data.” *Under Collaboration review; planned submission to Physical Review D.*

Refereed Journal Publications (First/Lead Author)

- **Seikh, M.F.H. (2025).** “A geometrical approach to neutrino oscillation parameters”. *AIP Advances* 15 (10), 105204. DOI:10.1063/5.0292233.

- **Seikh, M.F.H., Grachev, V., Kravchenko, I., Athar, M.S., & Besson, D.** (2025). “First Radio Neutrino & Cosmic Rays Astronomy Workshop in India”. *Eur. Phys. J. Spec. Top.* 234(16), 4919–4924. arXiv:2509.20704; DOI:10.1140/epjs/s11734-025-01955-8.
- **Seikh, M.F.H.** (2025). “Physics of radio antennas”. *Eur. Phys. J. Spec. Top.* 234(16), 4967–4983. arXiv:2411.07507; DOI:10.1140/epjs/s11734-025-01500-7.
- **Seikh, M.F.H. for the ARA Collaboration** (2025). “Askaryan Radio Array: searching for the highest energy neutrinos”. *Eur. Phys. J. Spec. Top.* 234(16), 5065–5075. arXiv:2411.01761; DOI: 10.1140/epjs/s11734-025-01708-7.

Conference Proceedings (First/Lead Author)

- **Seikh, M.F.H., Jarvis, R., & Stiles, J.** (2026). “AAFIYA: Antenna Analysis in Frequency-domain for Impedance and Yield Assessment”. Accepted for the 2026 *IEEE AP-S/URSI Symposium*. arXiv:2601.13583.
- **Seikh, M.F.H., Giri, P. & Besson, D. for the ARA Collaboration** (2025). “Techniques for Continuous Wave Identification and Filtering in the Askaryan Radio Array”. PoS(ICRC2025)1171; arXiv: 2509.20735.
- **Seikh, M.F.H. & Besson, D. for the RNO-G Collaboration** (2025). “Observation of Radio Solar Flares in RNO-G”. PoS(ICRC2025)1365.
- **Seikh, M.F.H. for the ARA Collaboration** (2023). “Calibration and Physics with ARA Station 1”. PoS(ICRC2023)1163; arXiv:2308.07292.

Invited and General Scientific Writing

- **Seikh, M.F.H. & Besson, D.** (2026). “Polar Ice as a Laboratory for Physics”. *Physics Bulletin 2026, Physics Department, AMU*, pp. 27–37.
- **Seikh, M.F.H. & Besson, D.** (2024). “IceCube Neutrino Observatory: A Cold Discovery Hub”. *Physics Bulletin 2024, Physics Department, AMU*, pp. 14–23.
- **Seikh, M.F.H.** (2020). “Neutrinos and the 40 Future Routes”. *Physics Bulletin, Physics Department, AMU*. DOI:10.13140/RG.2.2.27981.10729.

Collaborative Works

Complete publication record, including contributions to the ARA, RNO-G, ARIANNA, IceCube, IceCube-Gen2, PUEO, TAROE-M, and RET Collaborations, available on:

Inspire HEP ; Google Scholar ; ResearchGate ; ORCID  ; Scopus

Professional Service

Editorial

- **Book Editor** for Springer’s EPJ-ST journal: “Radio Detection of Ultra-high Energy Neutrino & Cosmic Ray” (15 articles) with M.S. Athar, D. Besson, V. Grachev, and I. Kravchenko (View Cover) (2024–2025)
- **Founder and Editor-in-Chief** of *The Magnetic Monopole* e-magazine for KU’s Society of Physics Students (View Issue) (Jun 2022)

Research Article Review

- **Peer Reviewer:** for World Scientific’s IJMP-A journal (1 article, 2 reviews) (2025)
- **Peer Reviewer:** for Springer’s EPJ-ST journal (5 articles, 11 reviews) (2024–2025)
- **Internal Collaboration Reviewer:**
 - ✓ Muzio, M.S. and Salcedo-Gomez, A. for the ARA Collaboration, “The First Array-Wide Diffuse Flux Search for UHE Neutrinos with the Askaryan Radio Array”, 11th International Workshop on Acoustic

- and Radio EeV Neutrino Detection Activities (ARENA2026) (*Jun 2026*)
- ✓ Giri, P. for the ARA Collaboration, “A Comprehensive Diffuse Neutrino Search Using the Full Askaryan Radio Array”, 32nd International Symposium on Lepton Photon Interactions at High Energies (Lepton-Photon2025) (*Nov 2025*)

Conference Session Chair

- **General Cosmology:** for APS Global Physics Summit (Mar 19, 2026)
- **Topical Groups Session II:** for APS Global Physics Summit (Mar 17, 2026)

Other Reviews

- **REU Application Reviewer:** for KU Physics and Astronomy (10 Applications) (2025)
- **SSGSA Application Reviewer:** for SSGSA (70+ Applications) (*2021–2026*)

Professional Leadership & Membership

- **American Physical Society (APS):** Member (*2020–Present*)
- **National Society of Black Physicists:** Member (*2021–Present*)
- **Physics & Astronomy Graduate Student Organization (GSO), KU:**
 - ✓ Graduate Students Representative - Member at Large (*2025–2026*)
 - ✓ Graduate Students Representative - Liaison (*2024–2025*)
 - ✓ GSO Executive Officer (*2023–2024*)
- **Sir Syed Global Scholar Award (SSGSA):**
 - ✓ Peer Mentor (*2023–Present*)
 - ✓ Team Lead, Promotions & Outreach (*2023–2025*)
- **Society of Physics Students, KU Chapter:** Graduate President (*2021–2022*)
- **Sigma Pi Sigma (National Physics Honor Society):** Lifetime Member
- **AMU Society of Physics:** Member (*2018–Present*)

Technical Proficiencies

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- **Programming:** Python, C/C++, Fortran
 - **Scientific Computing & Data Analysis:** CERN ROOT, uproot, NumPy, SciPy, pandas, scikit-learn
 - **Detector, Simulation & RF Software:** AraRoot/AraProc, AraSim, NuRadioMC, HFSS, Mathematica
 - **Development & Computing:** Git/GitHub, Linux (Ubuntu), macOS, Windows
 - **Document Preparation:** L^AT_EX, Microsoft Office Suite

Honors & Awards

Academic Honors

- **University Gold Medals:**
 - ✓ M.Sc. Physics (Top among 50 students), AMU (*2020*)
 - ✓ B.Sc. Honors Physics (Top among 120 students), AMU (*2018*)
 - ✓ Highest percentage in B.Sc. (Hons) Faculty of Science, AMU (*2018*)
- **Abdul Aziz Medal** for B.Sc. Physics, AMU (*2018*)
- **Undergraduate Topper Scholarship**, UGC India (*2020*)
- **University Merit Financial Award**, AMU (*2019–2020*)

Fellowships & Grants

- **Ralston Dissertation Fellowship**, KU Physics (*2025–2026*)
- **McKeithan Summer Fellowship**, KU Physics (*2025*)

- **Graduate Student Award for Distinguished Service**, KU (2024–2025)
- **Wallace S. Strobel Scholarship**, KU EECS (2024–2025)
- **Ralston Summer Fellowship**, KU Physics (2024)
- **Doctoral Student Research Fund**, KU (2024)
- **Distinguished Students Award**, APS FIP (2021–2022)
- **Carl & Catherine Chaffee Scholarship**, KU Physics (2020–2021)

Conference Travel Awards

- **Junior Participant Support**, Neutrino (2026)
- **Graduate Student Travel Fund**, KU Physics (2023, 2025)
- **DAP Travel Grant**, APS (2024)
- **DPF Travel Award**, APS (2022, 2024)
- **CLAS Graduate Scholarly Development Fund**, KU (2022)

Other Distinctions

- **Washington DC Scholarship**, AMU Alumni Association (2015–2019)
- **Sir Syed Global Scholar Award** for prospective graduate students (2018)
- **Senior Diploma in Painting with Distinction**, Bangiya Sangeet Parishad (2014)

Invited Lectures, Seminars, & Colloquia

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- **Searching for the Universe’s Highest-Energy Neutrinos with Radio Detectors**
Particle Physics Seminar, Carnegie Mellon University (*Sep 9, 2026*)
 - **Radio Engineering for Particle Astrophysics Experiments**
HEP Seminar, University of Nebraska-Lincoln (*Mar 26, 2025*)
 - **Radio Engineering for Particle Astrophysics Experiments**
Astroparticle Seminar, Ohio State University (*Jan 22, 2025*)
 - **Radio Engineering & Antenna Physics**
Lecture Series, IEI Summer School 2024, University of Alaska, Anchorage (*Jul 2–5, 2024*)
 - **UHE Neutrino & Cosmic Ray Detectors**
ICEHAP Seminar, Chiba University, Japan (*Aug 9, 2023*)
 - **Antenna Physics & Measurements**
Lecture Series, IEI Summer School 2023, South Dakota School of Mines (*Jul 2–5, 2023*)
 - **UHE Neutrino & Cosmic Ray Detectors**
Physics Colloquium, AMU, India (*Jan 12, 2023*)
 - **Antennas in UHE Neutrino Detectors**
Particle Physics Seminar, University of Delaware (*Oct 4, 2022*)

Conference Presentations

Oral Presentations

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- **Pioneering an Array-Wide Search for Ultra-high Energy Neutrinos with ARA**
APS Global Physics Summit (*Mar 18, 2026*)
 - **RNO-G in-ice Antennas**
RNO-G Collaboration Meeting, Uppsala University (*Sep 9–14, 2024*)
 - **Radio Frequency Instrumentation**
Radio Neutrino Workshop, AMU (*Apr 18–20, 2024*)
 - **Towards Five Station ARA Analysis**
APS April Meeting, Sacramento (*Apr 3–6, 2024*)

- **ARA Antenna Models**
ARA Collaboration Meeting, Ohio State University (*Mar 5–8, 2024*)
- **Current Status & Future of ARA**
APS Prairie Section Meeting, University of Missouri (*Nov 30–Dec 2, 2023*)
- **Introduction to Antenna Physics**
RET Collaboration Meeting, Vrije University Brussels (*Nov 8–11, 2023*)
- **Gain Calibration of RNO-G Antennas**
RNO-G Fall Meeting, Penn State University (*Sep 11–15, 2023*)
- **Galactic Noise in Radio Neutrino Experiments**
Astro Seminar, KU (*Oct 28, 2022*)
- **Towards ARA Station 1 Calibration**
APS April Meeting, New York City (*Apr 9–12, 2022*)
- **RNO-G Ice Calibration Planning**
RNO-G Collaboration Meeting, University of Chicago (*Apr 8, 2022*)

Poster Presentations

- **A Diffuse Search for UHE Cosmogenic Neutrinos with ARA Five Stations**
Neutrino 2026, UC Irvine (*Jun 21–26, 2026*)
- **Monitoring Snow Accumulation in ARA Using DAQ Temperature Data**
Neutrino 2026, UC Irvine (*Jun 21–26, 2026*)
- **A Geometrical Approach to Neutrino Oscillation Parameters**
APS Global Physics Summit (*Mar 17, 2026*)
- **CW Identification & Filtering Techniques in ARA**
39th ICRC, Geneva, Switzerland (*Jul 19, 2025*)
- **Observation of radio solar flares in RNO-G**
39th ICRC, Geneva, Switzerland (*Jul 15, 2025*)
- **Gain Measurements of LPDA Antennas**
APS April Meeting, Sacramento (*Apr 3–6, 2024*)
- **Overview of ARA Detector: Five Station Analysis**
ICTS Neutrino School, India (*Apr 22–May 3, 2024*)
- **Calibration of ARA Station 1**
38th ICRC, Nagoya University, Japan (*Jul 29, 2023*)

Outreach & Leadership

- Co-organized and hosted India’s inaugural workshop on **Radio Detection of UHE Neutrino and Cosmic Rays** at AMU (*Apr 2024*)
- Conducted outreach seminar on **Physics Research Opportunities** at AMU (*2021*)
- Participated in India’s first **Science Leadership Workshop** (*2020*)